



TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

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UNIT TEST - 1, JULY - 2024

Time Allowed : 1.30 Hours

PHYSICS

[Max. Marks : 35]

PART - I

Answer all the questions

5x1=5

- The Dimensional formula of Planck's constant
a) $[ML^2T^{-1}]$ b) $[ML^2T^{-3}]$ c) $[MLT^{-1}]$ d) $[ML^3T^{-3}]$
- Random Error can also be called as -----
a) Personal Error b) Chance Error c) Gross Error d) System Error
- If the particle has Negative Velocity and Negative Acceleration, its speed.
a) Increases b) Decreases c) Remain Same d) Zero
- Number of Significant digits in 2030.
a) 1 b) 2 c) 3 d) 4
- The ratio of the displacement vector to the corresponding time interval is
a) Average Speed b) Average Velocity
c) Instantaneous Speed d) Instantaneous Velocity

PART - II

Answer any three questions. (Question Number 10 is Compulsory)

3x2=6

- What are the Limitations of Dimensional Analysis?
- Check the dimensional correctness for the given Equations. $S = ut + \frac{1}{2}at^2$
- Define : Vector Give examples.
- Write down the Kinematic Equations for Angular Motion.
- Two vectors are given as $\vec{r} = 2\hat{i} + 3\hat{j} + 5\hat{k}$ and $\vec{F} = 3\hat{i} - 2\hat{j} + 4\hat{k}$. Find the resultant vector $\vec{\tau} = \vec{r} \times \vec{F}$

PART - III

III Answer any three of the following. Q.No.15 is Compulsory.

3x3=9

11. Write the rules for determining significant figures.
12. Mention the properties of Scalar.
13. Derive the relation between Linear Speed and Angular Speed.
14. How will you measure the diameter of the moon using Parallax method.
15. A object is thrown with Initial speed 5ms^{-1} with an Angle of Projection 30° . What is the height and range reached by the Particle?

PART - IV

IV Answer the following in detail:

3x5=15

16. a) Explain in detail the triangle Law of Addition.

(OR)

- b) Derive the expression for Centripetal Acceleration.

17. a) Explain in detail the various types of Errors.

(OR)

- b) Obtain an expression for the time period (T) of a Simple Pendulum. The time period T depends on m, l, g at the place where the Pendulum is suspended ($K = 2\pi$)

18. a) Write a note on Triangulation method and Radar method to measure Larger distance.

(OR)

- b) Derive the Kinematic equation of Motion for Constant Acceleration.



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